What to fix

System
supervisor_research_math

Runs from
Oct 2, 2026 (UTC)

Runs analyzed
20 runs

Models used by the agents
claude-haiku-4-5-20251001 (100 model calls)

Analysis cost
$1.13 for 165 AI judge answers (includes cached results, paid once)

20 of 20 runs had a serious problem. The most common: Supervisor can't see which agent replied (20 runs).

20 (100%)	$0.0049	4.7 s	7.5 steps	100%
Runs with a serious problem of 20 runs	Median cost per run	Median time per run	Median steps per run	Share of total cost spent in runs with a serious problem

Top three fixes

Only 2 problems qualified for this page: serious or moderate problems found with high or medium confidence, and cost or token problems only when they are at least 10% of the system's cost. Everything else is under Other findings on page 2 or in details.pdf.

1. Supervisor can't see which agent replied

Confidence: High, found by rule

Where	supervisor, final reply
What happened	In 20 of 20 runs. For example, in R1: supervisor runs on claude-haiku-4-5-20251001 and received research_expert's reply with its name only in the name attribute.
Why it matters	The supervisor's model sees the other agents' replies with no names, so it can mistake whose work it is reading, redo it, or drop it from the answer. It affected 20 of 20 runs (100%), which cost $0.12 and took 1.8 min in total.
What to change	Pass include_agent_name="inline" to create_supervisor, so supervisor's model sees each agent's name in the message text.
Evidence	Finding F1, details.pdf page 3

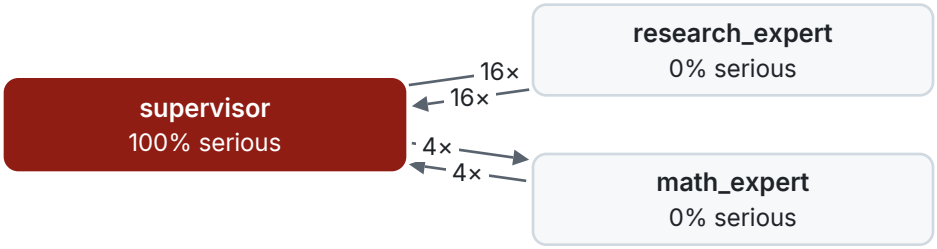
2. Agent ignored its instructions

Confidence: High, found by AI judge

Where	research_expert, reply 2
What happened	In 9 of 20 runs. For example, in R8: research_expert did math despite 'Do not do any math'.
Why it matters	Your system depends on each agent keeping to its role; when one doesn't, the result can't be trusted. It affected 9 of 20 runs (45%), which cost $0.0400 and took 40.2 s in total.
What to change	Add to research_expert's prompt: 'Only report the raw figures you found; never add, total or average them. Return the numbers to the supervisor so the math_agent can compute.'
Evidence	Finding F21, details.pdf page 5

Where and how often

Agents and handoffs



0% 100% of its runs

Boxes are agents and arrows are handoffs between them. Darker boxes had a serious problem in a larger share of the runs they took part in.

Agents

Agent	Runs	Runs with a problem here	Median cost	Median time	Most common problem
supervisor	20	20 (100%)	$0.0013	1.5 s	Supervisor can't see which agent replied
research_expert	16	9 (56%)	$0.0039	3.8 s	Agent ignored its instructions
math_expert	4	0 (0%)	$0.0028	2.1 s	none

Other findings

Problem	Agent	Runs affected
One agent used most of the tokens	research_expert	3 of 20 runs

2 more in details.pdf

What's working

- supervisor received what research_expert wrote in 16 of 20 runs.
- research_expert received what supervisor wrote in 3 of 20 runs.
- supervisor received what math_expert wrote in 2 of 20 runs.

How to read this report: details.pdf, section 1.